

Steps to Setup Raspberry Pi Video Looper for Vector (without WiFi)

YouTube videos: <https://youtu.be/HL4eKUpZwSk> (general approach)

<https://youtu.be/IEZOzp76d5s> (Vector-specific info)

<https://youtu.be/MsBr2YUHggI> (Better protection for video display!)

Step 1: You will need:

- Raspberry Pi Zero WH (H for GPIO Header already soldered!) \$23.20
- Waveshare 5" DPI LCD Screen (be sure to get DPI) \$49.95
- MicroSD card (Sandisk, 64 GB) \$13.49
- MicroSD card reader/writer \$9.98
- USB drive (64 GB) \$16.99
- Micro USB male to USB female converter (known as OTG, or on-the-go) cable \$4.99
- Raspberry Pi 5V 3A power supply \$17.49
- Electrical outlet, \$5.00 <https://www.homedepot.ca/product/legrand-wiremold-duplex-receptacle-box-ivory/1000132142>
- Computer with internet connection (not priced; I assume you have one!)

Total Cost: \$141.09 (Canadian, approximately \$100 US)

Other things you may want/need:

- Raspberry Pi Starter Kit: \$15.99
 - Case (two parts, top and bottom)
 - Mini HDMI to HDMI connector cable
 - Heat sink
 - 2x20 pin header (not needed if Zero WH used above)
 - Micro USB male to USB female converter OTG (on-the-go) cable
 - Power cable with switch (microUSB to microUSB)
- Power cable with switch (microUSB to USB) \$8.29
- USB power supply (or old phone charger. Should be 2.5A. Mine is 2.4A and works fine) \$12.39
- USB Male/Female Charging Extension Cable, 22AWG, 6 ft, \$13.99 (Pkg of 2)
- USB Female/Open End Cable, 22AWG, 30 cm, \$10.00 (Pkg of 2)
- Mini HDMI to HDMI connector cable \$7.99
- HDMI cable \$9.99
- USB Hub \$9.99
- USB Keyboard, mouse \$24.97

Step 2: Install the Operating System and Setup the Video Looper

Download operating system and looper software (based on DietPi/Buster):

- Go to <https://videolooper.de>, a German site with free video looping software
- Select latest version, click on it to download. Be patient! It will eventually download, but may take a long time to even start. Do not touch video_looper_v2.6.zip.download. It will soon be unzipped and replaced by video_looper_v2.6.img. This is a disk image of the operating system for the next step.
- Place MicroSD card in card reader, reader in USB port of your home computer
- Go to <https://www.raspberrypi.com/software/>
- Click on "Download for macOS" (or as appropriate). Wait for download.
- Click on downloaded file (Imager_1.8.4.dmg) to get Raspberry Pi Imager
- Double-Click on "Raspberry Pi Imager.app"
- Choose Device Raspberry Pi Zero, Choose OS Custom, video_looper_v2.6.img, Choose Storage Untitled volume (the MicroSD card), NEXT
- NO to apply customization, YES to erase existing data, password for your computer, wait...
- Done! MicroSD card is named boot

Changes required for best display of video:

- Edit the file /boot/video_looper.ini using textedit.app
- Find the line (near the end of the file) that begins "extra_args"
- Add the command "--win 0,-60,800,420 --aspect-mode letterbox" " to the end of this line:
- extra_args = --no-osd --audio_fifo 0.01 --video_fifo 0.01 --align center --font-size 55 --win 0,-60,800,420 --aspect-mode letterbox
 - This will display your video as ultrawide with black bars top and bottom, and scroll the top black bar off the top of the screen, in order to fill the upper half of the Waveshare screen.

Step 3: Install the Waveshare Display Software

Download and modify Waveshare software:

- Go to https://www.waveshare.com/wiki/Main_Page Waveshare display list
- Select LCD screen (5-inch DPI LCD), takes you to https://www.waveshare.com/wiki/5inch_LCD_for_Pi
- Follow instructions following "For Raspberry Pi OS Buster branch":
 - Place MicroSD card in card reader, reader in USB port of your home computer
 - Use text editor to edit config.txt (comment out one of [pi4] or [all] optional lines):
 - Add the following lines at end of file:
 - # Enable Waveshare 5-inch DPI LCD screen (Buster OS)
 - # [pi4]
 - # dpi_timings=800 0 40 48 88 480 0 13 3 32 0 0 0 60 0 26000000 6
 -
 - # [all]
 - dtoverlay=dpi24
 - enable_dpi_lcd=1
 - display_default_lcd=1
 - dpi_group=2
 - dpi_mode=87
 - dpi_output_format=0x6f005
 - hdmi_timings=800 0 40 48 88 480 0 13 3 32 0 0 0 60 0 32000000 6
 - disable_overscan=1
 - PLUS THESE ADDITIONAL LINES TO ROTATE SCREEN 180 DEGREES:
 - # Rotate by 90 degrees times value (e.g., 90 x 3 = 270 degrees)
 - display_rotate=2
 - Save file. Mine said I didn't have permission, but somehow saved anyway. If problems, try closing without saving, then open to see if it actually saved. If you still have problems, try Duplicating the file, then rename the two files.
 - Save and quit the MicroSD card safely, remove from home computer

Step 4: Setup and Run the Raspberry Pi with the Waveshare Screen

- Insert the MicroSD card into the Raspberry Pi
- Connect the gold cable on the WaveShare 5-inch to the JP3 connector on the small blue circuit board, by carefully pulling the black clip on the JP3 out about 1/8" (both ends), sliding the gold cable in, connector side down, and then sliding the black clip back into place.
- connect the large blue circuit board to the 40-pin GPIO interface of the Raspberry Pi. Don't get it backwards! The Pi and the board should make a nice little cube. Make sure the LCD backlight switch is set to ON.
- Plug USB adapter cable (micro-USB to USB) into **inner** micro-USB connector.
- Plug USB drive with at least one video file in the root directory into the USB adapter
- Plug power cable (micro-USB to USB) into **outer** micro-USB connector. Plug into power converter / phone charger. Should be 5 Volt and minimum 2.4 Amp converter. Plug this into wall outlet.
- Click power button on your power cable, if there is one, to turn on Raspberry Pi. It will take a few seconds to start.
- See that blinking cursor in the top left of the screen? That means it's working.
- Messages should begin scrolling up screen. Just ignore.
- Eventually you should see a countdown starting at 5, followed by your videos. If you see a message, "Insert USB drive with compatible movies", there is a problem with your USB drive. Check that you have compatible video formats (.avi, .mov, .mkv, .mp4, or .m4v), and they are in the root directory.

Step 5: If You've had Problems

- Check the Waveshare.com and videolooper.de websites (where you got downloads above) for additional information. DietPi.com has information on the operating system:
 - <https://videolooper.de> Help with getting the looper to work
 - https://www.waveshare.com/wiki/5inch_LCD_for_Pi Help with getting the screen to display
 - <https://dietpi.com/docs/> Help with the operating system
 - <https://github.com/popcornmix/omxplayer#synopsis> Help with the video player

Step 6: Logging onto the Raspberry Pi and Possible Changes

- Remove USB drive from Raspberry Pi, and attach keyboard
- Hit <Esc> when it displays "Insert USB drive with compatible movies"
- Login ID is pi, password is video (you may need to hit <Enter> first to get the login prompt).
- Change password for pi (good safety measure):
 - Type passwd, and follow the instructions
- Update time zone and keyboard (should be set from Imager, step 2):
 - Type sudo dietpi-config
 - Use the four arrow keys on the keyboard to move around the screen
 - Language/Regional Options (should have been set from step 2):
 - Change time zone to America/Toronto (or as appropriate)
 - Change to US keyboard (or as appropriate)
- Edit basic Raspberry Pi settings:
 - Sudo nano /boot/config.txt
 - Make edits as required
 - <Ctrl-X> to exit, <Y>es, <Enter> to save over old file
- Edit video looper instructions:
 - Sudo nano /boot/video_looper.ini
 - Make edits as required
 - Extra_args is very useful to fix video issues. It sends these commands to the OMX video player. Check out <https://github.com/popcornmix/omxplayer#synopsis> for additional useful OMXPlayer instructions. I could not get --win or --crop to work, but --aspect-mode worked to fill the screen, instead of black borders top and bottom.
 - <Ctrl-X> to exit, <Y>es, <Enter> to accept save filename
- Add User: Sudo adduser dave
- Remove User: sudo userdel -r dave
- List Users: getent passwd {1000..6000}
- Change Hostname: sudo nano /etc/hostname, change RaspberryPi to Vector, save (ctrl-X, Y, Enter)
- Shutdown: sudo halt
- Reboot: sudo reboot

Step 7: Edit Your Video

- By default, videos will run in alphabetical order. Rename files starting with a number to control the order.
- There is a black gap between clips. To avoid this, it's best to combine all clips into a single file.
- If your pinball is movie-themed, there are likely video clips from the film available on YouTube:
 - Go to YouTube
 - Search "Rollerball movie clips" (or appropriate film title)
 - Select a clip
 - Download it to your computer:
 - Pause the clip, and click on the top line URL (e.g., <https://www.youtube.com/watch...>)
 - Change the word youtube to ssyoutube, and hit enter
 - This takes you to a download screen. Click on "Download MP4 720" when it is displayed.
 - NOTE: This download method no longer appears to work. Try Googling "How to download videos from YouTube" for a current method.
 - Edit your clips to display on the top 1.5 inches of screen only, as the screen will be half-hidden behind the HYPE targets.
 - Edit clips
 - Use your favourite video editor to cut down and combine clips, keeping only the parts you want
 - The `extra_args` command we modified above ensures the top of your video will be displayed at the top of the screen.
 - UltraWide videos are perfect for the 5-inch screen, filling exactly the area above the HYPE targets. Widescreen videos are more difficult. When you reduce them to fill the top 1.5 inches of the screen, they leave a large part of the visible screen unused. I found this remaining screen was exactly the right size to fit in three minimized widescreen videos, one on top of the other. The trick is to find appropriate videos to fill the area! I used a clip of the Addams Family (for reasons obvious to Vector owners), old commercials, other old TV shows, comedy sports events, etc.
- Copy edited clip files to your USB drive, and attach it to the Raspberry Pi with the connector cable

Step 8: Install the Raspberry Pi in Your Pinball Machine

- Remove the announcer plastic from behind the HYPE targets. Remove reflective metal HYPE roof and blue posts holding it (held by bolts rather than screws!)
- Replace two #555 bulbs with LEDs, to eliminate heat issues between bulb and cable using same 1" hole! You can also cut a thin piece of Styrofoam to fit in the hole and insulate the cable from the bulb.
- You need to make more room behind hype targets for the screen! Replace two blue posts behind hype targets with left-most two metal posts from under announcer plastic. Blue posts can be used to hold announcer. Wrap static foam insulation around metal posts. The left post should be wrapped in two strips, above and below indent, to allow space for the rubber. Wrap the foam insulation with duct tape. This makes more room, and cushions the screen.
- Add a new blue post to the right side of the hype targets, to protect right side of screen, and to take up the excess length of the 3½" rubber. See photo below.
- Use 3½" rubber around display and posts. Display should be held against foam-wrapped metal posts. Rubber should be behind hype targets, and under Vector plastic behind display. Slide display to right, so it is covered by half of left post. This maximizes protection on the left, where it is most needed.
- Cut a thin piece of black plastic to same size and shape as reflective metal HYPE roof. Use this to replace the reflective surface. Trim this at the back, to leave as much room for the display as possible.
- Screw posts back into place, with black plastic roof on top.
- Disconnect screen from Raspberry Pi by releasing black clips (pull out approx. 1/16") and removing white ribbon cable.
- Install Raspberry Pi under the playfield. Check how ribbon cable will fit through 1" hole and how it will reattach to display, to make sure you won't have to twist it! Insert two screw eye hooks into bottom of playfield near the 1" hole. Attach Raspberry Pi to screw eyes with zip ties so that it hangs freely.
- Work white cable up through the 1" lamp hole. Reattach to display and close the black clips.
- Put the announcer plastic in place, and screw it down.
- There are two ways I can think of to power your setup. One is to run a USB extension cable up into the backbox, connecting it to a USB connector that you have attached to an available 5-volt and ground source. I have not done this and so I can't advise you much more than this. The other way to power the setup is to install a power outlet in the cabinet by tapping into the 120-volt power lines running from the power switch on your pinball to the transformer. Both of these have their challenges.
- Unplug the machine before doing any work on the power lines!
- To install a power outlet: Cut the blue and yellow, 120-volt power lines running from the power switch to the transformer. Connect the two cut ends of the blue line to the silver connectors on the outlet, and the two ends of the yellow lines to the gold connectors. Attach outlet to side of cabinet, and add cover.

- When adding a new power outlet in the cabinet, place it on the side, rather than the bottom, so dirt and solder don't fall into it and cause problems. Close the unused outlet with a child safety plug. The sides are 3/4-inch plywood, so use screws less than that!



Step 9: Keep copies of everything you downloaded for this project, in case it is no longer available online when you need it!

Keep copies of:

- Imager_1.8.4.dmg
- Video_looper_v2.6.img
- Instructions above, including lines added to config.txt and video_looper.ini
- Any additional changes you did for your particular setup
- All your video files

Keep a copy on your home computer, and include it in your regular backup procedures.

Keep a copy on the USB drive, and maybe even the MicroSD card.

Take screen shots of these instructions. Print them out and keep them with your pinball manual!

The next owner of your pinball will thank you.

Pros and Cons of Powering off the Raspberry Pi

(As opposed to using a shutdown command)

There is an issue with turning the Raspberry Pi off without shutting it down first. Just like shutting down your computer, you don't just unplug it. Theoretically you could interrupt some important file maintenance operation and corrupt your MicroSD card. But, worst case scenario, your Pi will stop running, so you pull out the MicroSD card and repeat all the steps above.

A better way to shut it down would be to connect it to your WIFI, and get an app on your phone that can shut it down remotely. Theoretically this can be done, but I didn't get it working. I was able to get WIFI working with other operating systems, but couldn't get video looping working with any of them!

Cons

- If Pi is interrupted while writing to MicroSD, you can corrupt it, and videos will no longer run
- I will be giving up on resolving this issue. And that really bugs me.

Pros

- Very simple
- No dependency on human intervention
- No one but me will ever do anything else
- Will not feel bad when I don't bother
- Other commercial options handle it the same way. I am told there have been few problems (but not zero!)
- No WIFI connection needed
 - WIFI connection is a potential security risk
 - WIFI password not used, so no one can possibly hack your Pi remotely to access your network
 - I have not been able to get WIFI to work with this OS!
- The worst that can happen is, the MicroSD card needs to be reprogrammed by repeating all the previous steps
 - Absolute worst, maybe MicroSD needs to be replaced
 - ABSOLUTE worst, you may need to watch this video again

UPDATE: I have had the Raspberry Pi turn off and on with the pinball machine main switch, as setup in the description above, for a couple of years now. I generally try to turn it off while the video is running, not when it is booting up or between videos. I feel this minimizes any chance that the Pi is writing to the data card when it is shut down. So far it has given me no problems.